```
from google.colab import files
uploaded = files.upload()
```

Choose files Screenshot … 142020.png

**Screenshot 2026-07-13 142020.png**(image/png) - 268920 bytes, last modified: 13/07/2026 - 100% done

```
Saving Screenshot 2026-07-13 142020.png to Screenshot 2026-07-13 142020 (1).png
```

```
import cv2
import matplotlib.pyplot as plt

filename = list(uploaded.keys())[0]
img = cv2.imread(filename, 0)

# Blur image
blur = cv2.GaussianBlur(img, (5, 5), 0)

# High-Boost filtering
A = 2
sharp = cv2.addWeighted(img, A, blur, -(A - 1), 0)

plt.imshow(sharp, cmap='gray')
plt.title('High-Boost Sharpening')
plt.axis('off')
plt.show()
```

High-Boost Sharpening

```
Successfully downloaded input.jpg
```


```
Cause of Aliasing:
- Aliasing occurs when the image is sampled below the Nyquist rate.
- High-frequency details appear as jagged edges or moire patterns.
- Quantization reduces pixel intensity precision but is not the main cause of

Corrective Approach:
- Apply an anti-aliasing (Gaussian) filter before downsampling.
- Use better interpolation methods such as INTER_LINEAR or INTER_AREA.
- Increase the sampling resolution if possible.
```